3 Fertility

INTRODUCTION

Fertility refers to the actual production of children, which in the strictest sense is a biological process. A **zygote** is produced when the sperm of a male and the egg of a female are united, and around nine months later a baby is born. Most often in this process, though not always, a man and a woman have sexual intercourse, the woman conceives, and the **conception** results in a live birth. Even though the production of a child is a biological process, the various activities and events that lead to the act of sexual intercourse and, later, to giving birth are affected by the social, economic, cultural, and psychological characteristics of the woman and the man, as well as by the environment in which they live. The key to this seeming paradox is that engaging in intercourse, conceiving, and giving birth are themselves behaviors that are influenced by other factors, most of them social and cultural. So while we have no influence at all with regard to the family and parents we receive when we are born, we do have a significant influence on our own fertility, that is, whether or not we produce children, and if so, the number and timing of the children produced. That is, whether we decide to engage in sexual intercourse, whether this intercourse results in a conception, and whether a live birth is the outcome are all driven largely by social and cultural considerations.

Fertility can be studied in different ways, one of which is cross-sectionally, that is, at one point in time; a cross-sectional perspective is also known as a **period perspective.** Were we to study the fertility behavior of women and men in the year 2009, we would develop cross-sectional fertility measures (also called period measures) that would show the number of births to women and men in the calendar year 2009. Most of the fertility measures shown in this chapter are period measures; that is, they refer to a particular time period. A **period rate**, also called a cross-sectional rate, is a rate based on behavior occurring at a particular point or period in time.

Conversely, fertility may be studied over time to give us measures revealing the number and spacing of births to cohorts of women as they pass through the life cycle; this is known as **cohort analysis**. Here we could take the cohort of women who began their childbearing years at age 15 in, say, 1970. We could then follow them each year through 2005 when they were at the end of age 49 and had completed their childbearing years in order to see how many babies they had produced. Fertility may thus be measured on a cohort basis, as well as on a period basis.

The demographic study of the fertility of individual women (and men) is known as *microfertility analysis* because it refers to the fertility of persons. There are several different ways to study fertility at the individual level: 1) examining the number of births a woman (or man) has produced by a given point in time, such as the date of a census or survey; 2) examining the number of births a woman (or man) has had by the end of the childbearing years; and 3) focusing on the timing and spacing of births at various stages of the life cycle (say, between the ages of 25 and 29, or between 45 and 49).

Another way to study fertility is to use a *macro-level approach*, that is, to determine the rate at which births occur in a population or subpopulation during a given period of time. Rather than studying the fertility of persons, macrofertility analysis studies the fertility of populations (Poston and Frisbie, 2005). One reason demographers measure fertility at the macro level is to then compare it with mortality, and to compute rates of reproductive change. They also compare the fertility levels of different types of subpopulations over time.

In this chapter, we first consider the conceptualization and measurement of fertility. Second, we discuss the so-called **proximate determinants** of fertility. These are the mainly biological factors that lead directly to fertility, and are themselves influenced heavily by social factors. Third, we look at some of the main theories generated by demographers to specify the reasons that some women or men or societies have more babies than other women or men or societies. We next consider world fertility patterns and how they have changed over time, then focus on fertility trends and differences in the United States. We follow with a discussion of adolescent fertility, concluding with a section on male fertility.

CONCEPTUALIZATION AND MEASUREMENT OF FERTILITY

There are three main fertility concepts. **Fertility** is the actual production of male and female births and refers to real behavior. **Reproduction** is also actual production, but refers to the production of only female births (there is no demographic term to refer to the production of only male births). **Fecundity** refers to the potential or the biological capacity of producing live births.

The **crude birth rate (CBR)** is the first measure of fertility we consider. It is a cross-sectional (i.e., period) measure and refers to the number of births occurring in a population in a year per 1,000 persons. It is calculated as follows:

$$\text{CBR} = \frac{\text{number of births}}{\text{midyear population}} * 1{,}000 \qquad (3.1)$$

In 2007, the CBR for the world was 21/1,000. This means that in the world in 2007, there were 21 births for every 1,000 members of the population. Among the continents, the CBR in 2007 ranged from a high of 38 in Africa to a low of 10 in Europe. Almost four times as many children per 1,000 population were born in Africa than in Europe in 2007. North America had a CBR of 14 (the CBR of the United States was also 14); Latin America, 21; Asia, 19; and Oceania, 18 (Population Reference Bureau, 2007a). The major countries of the world had 2007 CBRs ranging from lows of 8 in Macao, South Korea, Germany, and Taiwan to highs of 50 in Liberia and the Democratic Republic of the Congo and 49 in Angola. Generally, CBRs above 30 are considered to be high, and those less than 15 to be low.

The CBR is referred to as "crude" because its denominator, the midyear population of the area, includes many people who are not at the risk of childbearing, such as young women (under age 15) and post-menopausal women (older than age 50). (An **at-risk population** is the population that is at the risk of the event of interest occurring to them.) Another downside is that men are included in the denominator, and, strictly speaking, men do not bear children, so are thus not exposed to the risk of childbearing. However, some demographers do study male fertility, and we consider this topic later in the chapter.

The **general fertility rate (GFR)** is another cross-sectional measure of fertility. It is superior to the CBR because it restricts the denominator to women of childbearing ages. The GFR is calculated as follows:

$$\text{GFR} = \frac{\text{births}}{\text{midyear population}_{\text{f},15-49}} * 1{,}000 \qquad (3.2)$$

where the numerator is the number of births in the population in the year, and the denominator is the number of females in the midyear population who are in the childbearing ages 15–49.

Table 3.1. Fertility data and rates for the United States in 2005

Age group Col. 1	Women in age group (midyear population) Col. 2	Live births to women in age group Col. 3	Age-specific fertility rates live births per 1,000 women ASFR Col. 4	ASFR × 5 Col. 5
15–19	10,240,239	414,406	40.47	202.3
20–24	10,150,079	1,040,399	102.50	512.5
25–29	9,767,524	1,132,293	115.92	579.6
30–34	9,906,365	952,013	96.10	480.5
35–39	10,427,161	483,401	46.36	231.8
40–44	11,475,863	104,644	9.12	45.6
45–49*	11,372,141	6,546	0.58	2.9
TOTALS	73,339,372	4,133,702		2,055.2[a]

* Data are for live births to mothers 45+ per 1,000 women 45–49.

[a] TFR = $\sum(\text{ASFR} * 5) = 2{,}055.2$.

Sources: For population data: U.S. Bureau of the Census, 2008; for birth data: Hamilton, Martin, and Ventura, 2007.

In the United States in 2005, there were 73,339,372 women of ages 15 to 49; and 4,133,702 babies were born in 2005 (see Table 3.1). Dividing the latter figure by the former and multiplying the result by 1,000 yields a GFR value for the United States in 2005 of 56.4. This means that there were more than 56 babies born in the United States in 2005 for every 1,000 women between the ages of 15 and 49.

Sometimes the denominator of the GFR is restricted to women between the ages of 15 and 44. This occurs because, as will be noted, not many babies are born to women in the 45–49 age group. To illustrate, of the 4,133,702 births that occurred in the United States in 2005, only 6,546, or 0.15 percent, occurred to women over the age of 44 (see Table 3.1).

In Figure 3.1, we show GFRs for the United States for individual years between 1970 and 2007. It is important to keep in mind when viewing these GFRs that the denominator is women 15–44, not women 15–49.

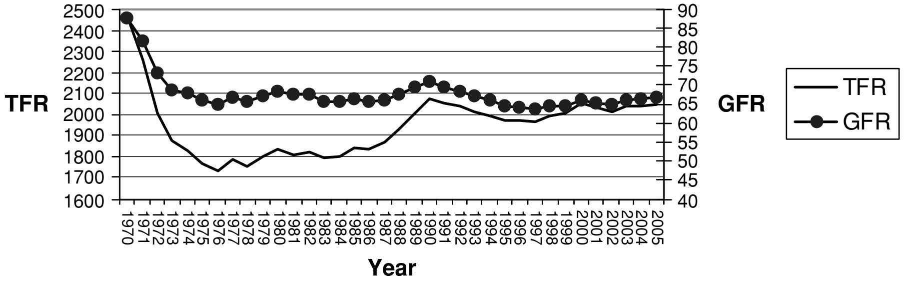

Figure 3.1. TFRs and GFRs: United States, 1970–2005. *Source:* The authors.

The data in the figure show that fertility as measured by the GFR (with a denominator of women 15–44) was high in 1970 but dropped to the 60s by 1980 and has mostly remained in the 60s since then. (Total fertility rates are also shown in Figure 3.1; they are defined later.)

In the definition for the GFR, we referred to its denominator as females in the childbearing ages. We need to consider here what is meant by *childbearing ages*. In practice, as already noted, few women give birth before age 15 and after age 49, and so demographers usually use the age range 15–49 to mark the limits of the childbearing ages. But this range is not universal. In developed countries, not much fertility occurs to women after age 44. Therefore, sometimes (see Figure 3.1) GFRs are calculated only for women of ages 15–44.

Of course, there are exceptions even to these age ranges. To illustrate, in the year 2000, the ages of mothers listed on birth certificates filed in Texas ranged from a low of 11 to a high of 53. The mean age of Texas mothers giving birth in 2000 was 26.1 years. Only 0.3 percent of all births were to mothers under age 15 and 1.6 percent to mothers of age 40 or older. More than half (55.0 percent) of all Texas births were to mothers 20–29 years of age and almost three-fourths (74.4 percent) were to mothers 20–34.

Percentages of U.S. births in 2005 to very young females are even lower than those for Texas. Less than 0.2 percent of all U.S. births in 2005 were to females ages 10–14. More than 2.4 percent of U.S. births in 2005 were to women older than the age of 40. Not many women under age 15 or over age 40, particularly older than age 45, have babies, but some do. Thus, although the denominator of the GFR starts at age 15, the numerator literally comprises of all births, including those to females under age 15.

In July 2008, the British Broadcasting Corporation (BBC) News agency distributed a story about, apparently, the oldest woman ever to give birth. In northern India, Omkari Panwar, 70 years old, gave birth to twins in 2008. She and her husband, Charam Singh, a farmer in his mid-70s, already had two children, both girls. They badly wanted a male, so they took out a bank loan to pay the costs for in vitro fertility therapy, the result being a boy and a girl, both weighing around two pounds (British Broadcasting Corporation News, 2008).

If one has data available only for the CBR but wishes to approximate the value of the GFR, an estimated GFR value (for women 15–44) is given by the following formula:

$$\text{GFR} = \text{CBR} * 4.5 \tag{3.3}$$

To illustrate, CBR values in the United States in 1950 and in 2005 were 24.1 and 14.0, respectively. If we had no other data available and needed GFR values for these two years, we could multiply each CBR by 4.5 and arrive at estimated GFR values of 108.5 for 1950 and 63.0 for 2005. These estimated GFRs are not far from the actual GFRs of 106.2 for 1950 and 66.7 for 2005. The constant in this formula and those in other formulas to be shown later are based on empirical and analytic relationships between the fertility measures (see Bogue and Palmore [1964] for an example of such an application).

We noted earlier that the GFR addressed the major problem of the CBR by restricting the denominator to women in the childbearing ages. A problem still remains, however, with the GFR. It does not take into account the fact that within the range of the childbearing years for females of 15 to 49, there are differences in the extent to which the women produce children. Fertility is usually low for women 15–19 and is then at its highest for women 20–29; the rates become lower in the 30s and even lower in the 40s. To take into account the fact that fertility varies by age, demographers calculate fertility rates for specific age groups of women.

The **age-specific fertility rate (ASFR)** reflects exactly what its name indicates: It focuses on births to women according to their age. ASFRs are usually calculated for women in each of the seven 5-year age groups of 15–19, 20–24, 25–29, 30–34, 35–39, 40–44, and 45–49. The general formula for the ASFR for women in age group x to x + n is calculated as follows:

$$\mathrm{ASFR}_{\mathrm{x\ to\ x+n}} = \frac{\mathrm{births}_{\mathrm{x\ to\ x+n}}}{\mathrm{females}_{\mathrm{x\ to\ x+n}}} * 1{,}000 \qquad (3.4)$$

Although most demographic analyses using ASFRs calculate them for these 5-year age groups, sometimes thirty-five single-year age groups are used, for example, age group 15, age group 16, age group 17, all the way up to age group 49.

Table 3.1 shows the numbers of women in the United States in 2005 in each of the seven age groups 15–19 to 45–49 (col. 2) and the numbers of babies born to the women in each of the age groups (col. 3). There were 10,150,079 women in 2005 in the age group 20–24. There were 1,040,399 babies born in 2005 to women in this age group. Dividing the latter figure by the former and multiplying the result by 1,000 produces an ASFR for women 20–24 of 102.5. This means that in the United States in 2005, for every 1,000 women 20–24, 102.5 babies were born to them. All seven ASFRs for U.S. women in 2005 are shown in Table 3.1. The highest ASFR is for women 25–29, the next highest for women 20–24, and the third highest for women 30–34. ASFRs for women in the other four age groups are not as high.

When the seven ASFRs are plotted, they usually form an inverted U. Such a plot is referred to as the *age curve of fertility*. Figure 3.2 shows six age curve of fertility plots for Africa, Asia, Europe, Latin America, North

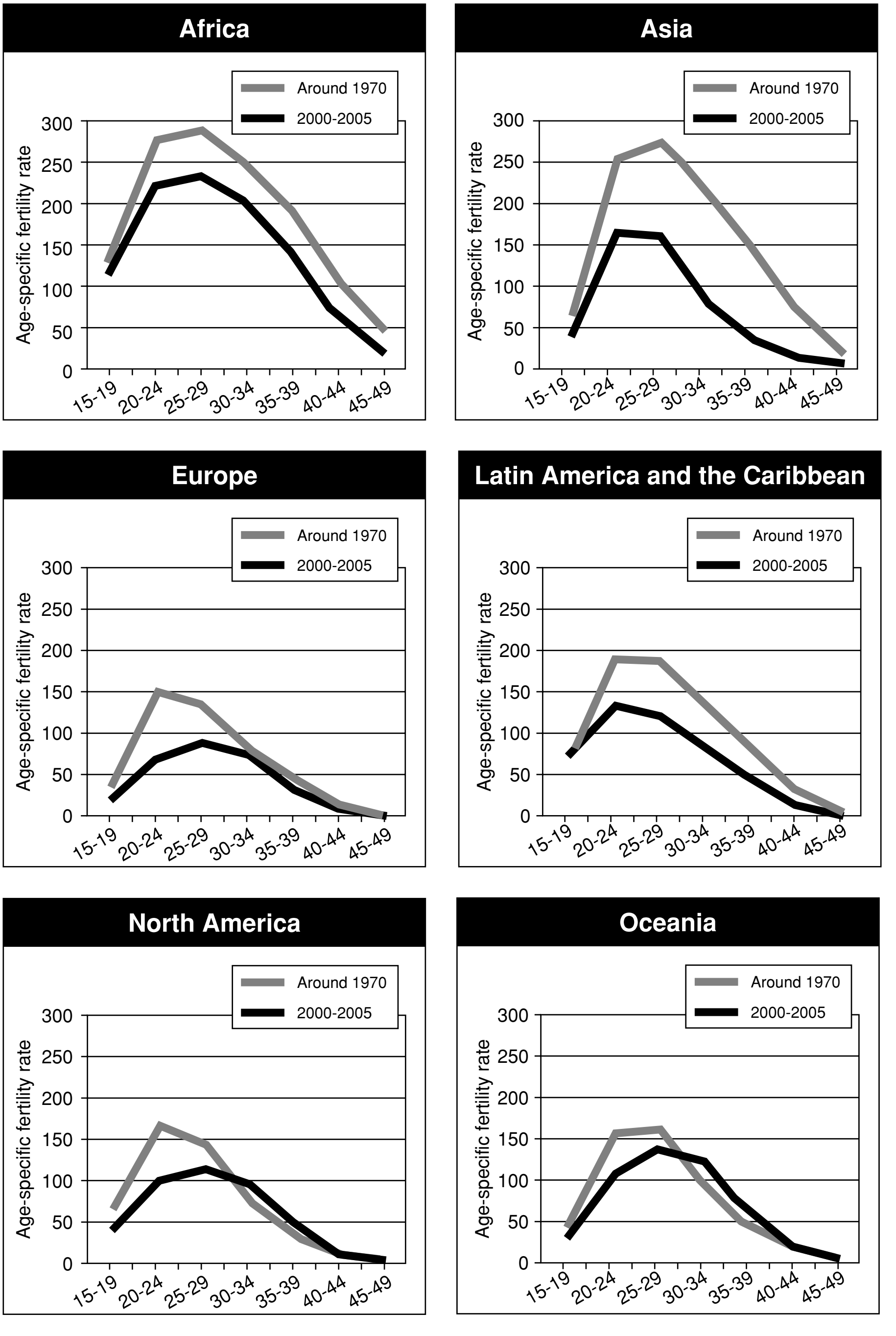

Figure 3.2. Trends in age-specific fertility, regions of the world: 1970 and 2000–2005. *Source:* United Nations, 2008a.

America, and Oceania; the curves for each region are shown for 1970 and for 2000–2005. Fertility as measured by ASFRs in 1970, as well as 2000–2005, is highest in Africa and lowest in Europe and North America. For the most part, fertility was higher in 1970 than 2000–2005. Later in this chapter, we focus on fertility and fertility change in the world.

The **total fertility rate (TFR)** is the most popular of all fertility rates used by demographers. Like the ASFRs, the TFRs take into account the fact that fertility varies by age; unlike the ASFRs, which are expressed quantitatively as a series of **specific rates** (usually seven, one for each age group), the TFR provides a single fertility value. The TFR is most frequently calculated cross-sectionally, that is, for a specific period of time, although as we will show, it may also be calculated for cohorts. A cross-sectional TFR for a particular point in time is an estimate of the number of births that a hypothetical group of 1,000 women would have during their reproductive lifetime, that is, between the ages of 15 and 49 (or, sometimes, 15 and 44), if their childbearing at each of their reproductive years followed the ASFRs for a given period. This number of live births that the hypothetical group of 1,000 women would produce if they were exposed to a particular schedule of ASFRs assumes that none of them die during their reproductive years.

The TFR is calculated by summing the ASFRs after multiplying each by the width of the age interval of the ASFRs. If we use ASFRs based on 5-year intervals, as is usually the case, we would multiply each ASFR by the value of 5:

$$\mathrm{TFR} = \sum (\mathrm{ASFR}_{\mathrm{x\ to\ x+n}} * \mathrm{i}) \tag{3.5}$$

where i = the width in years of the age interval; in most cases, this will be 5.

In Table 3.1, we show the calculation of the TFR for the United States for the year 2005. The ASFRs shown in column 4 are multiplied by the constant of 5 and reported in column 5. These values are then summed, to yield a TFR of 2,055. This may be interpreted as follows: If 1,000 women went through their reproductive years of 15 to 49 and were subjected each year to the U.S. ASFRs for the year 2005, by the time they reached age 49, they would have produced 2,055 babies, or an average of 2.1 babies each. The TFR is a standardized rate; that is, it is not influenced by the differences in the numbers of women in each age. Its value is especially useful in interpreting the fertility that is implied by a given set of ASFRs for a particular point in time.

TFRs are shown in Figure 3.1 for the United States for the years 1970 to 2005. In 1970, the TFR was almost 2.5, dropped to lows of 1.73 and 1.75 in 1976 and 1978, and then increased to its 2007 value of just slightly more than 2. There is a fairly close correspondence between the TFR and the GFR for most, but not all, of the years.

If one only has CBR data available for an area or country or, alternately, only GFR data, the TFR may be estimated with the following formulas:

$$\text{TFR} = \text{CBR} * 4.5 * 30 \tag{3.6}$$

$$\text{TFR} = \text{GFR} * 30 \tag{3.7}$$

For example, the actual CBR and GFR values for the United States in 2005 were 14.0 and 66.7, respectively. If we only had the CBR data but needed an estimate of the TFR for the United States in 2005, we could use formula 3.6 and obtain an estimated TFR of 1,890. If we only had available the GFR data and desired a TFR estimate, we could use formula 3.7 and get an estimated TFR of 2,001. These two estimated TFR values of 1,890 and 2,001 are fairly close to the true value of 2,055 shown in Table 3.1. Certainly, one would prefer to have the real data, but these two formulas are helpful for providing estimates.

In 2007, the world TFR was 2.7. It was 2.9 for the **less-developed countries (LDCs)** of the world including China, and 3.3 for the LDCs excluding China. The developed countries had an average TFR in 2007 of 1.6. TFRs ranged from a high of 7.1 in Niger and Guinea-Bissau to a low of 1.1 in South Korea and Taiwan (Population Reference Bureau, 2007a). If during their childbearing years, 1,000 women followed the ASFRs for Niger or Guinea-Bissau in 2007, at the completion of their 49th year, they would have produced 7,100 babies, or an average of 7.1 each. In contrast, if during their reproductive lives, 1,000 women were subjected to the ASFRs of either Taiwan or South Korea for the year 2007, by the time they ended their childbearing years, they would have given birth to 1,100 babies, or about 1.1 each. These are tremendous differences in fertility. We discuss TFR trends and differences among the countries of the world in a later section of this chapter.

We noted that the TFR may be calculated cross-sectionally, that is, for a specific period or year, and is referred to as a period or cross-sectional TFR. Alternately, the TFR may be calculated for cohorts. Rather than subjecting a hypothetical group of 1,000 women to a schedule of ASFRs for a given point in time, that is, as with the cross-sectional TFR, the cohort TFR follows a real group of women through their childbearing years and tabulates their actual fertility as they pass through these years. Table 3.2 presents average annual ASFRs for the years 1970–1974, 1975–1979, and so forth, through 2000–2004. Let us use these data to calculate a period TFR and a cohort TFR.

Table 3.2. Age-specific fertility rates, United States, 1970–1974 to 2000–2004

	15–19 years	20–24 years	25–29 years	30–34 years	35–39 years	40–44 years	45–49* years	Total fertility rate
1970–1974	62.3	137.1	124.1	61.9	25.5	6.2	0.4	2,087.5
1975–1979	53.0	111.8	109.1	112.4	19.2	4.2	0.2	2,049.5
1980–1984	51.9	110.7	110.5	63.9	21.2	3.9	0.2	1,811.5
1985–1989	52.4	109.5	112.9	72.7	26.5	11.3	0.2	1,927.5
1990–1994	59.8	113.2	115.5	80.0	32.4	5.9	0.3	2,035.5
1995–1999	52.0	107.8	109.4	83.7	35.9	7.1	0.4	1,981.5
2000–2004	43.7	104.8	114.3	93.0	42.2	8.4	0.5	2,034.5

* Before 1997, data are for 45–49 per 1,000 women; beginning with 1997, rates are for live births to mothers 45–54 per 1,000 women 45–49.

Source: The authors.

The cross-sectional TFR may be calculated for any of these years by summing the ASFRs across the respective row and multiplying the sum by 5. To illustrate, if we wanted the average annual cross-sectional TFR for the period 1970–1974, we would sum the seven ASFRs for that period, namely, 62.3, 137.1, 124.1, 61.9, 25.5, 6.2, and 0.4, and then multiply the total by 5 to produce an annual period TFR for 1970–1974 of 2,087.5 This value reflects the number of children that a hypothetical group of 1,000 women would produce were they to be subjected to the ASFRs of the United States for the 1970–1974 period.

In contrast, if we desired the cohort TFR for those women who started their childbearing years in 1970–1974 (we refer to those women as the 1970–1974 fertility cohort because they initiated their childbearing during those years), we would take the ASFRs on the diagonal line in Table 3.2 and observe the ASFRs for the 1970–1974 fertility cohort of women as they proceeded through their childbearing years. When those women were 15–19 in 1970–1974, they had an ASFR of 62.3; when they were 20–24 in 1975–1979, they had an ASFR of 111.8; when they were 25–29 in 1980–1984, they had an ASFR of 110.5; and so forth until they were 45–49 in 2000–2004 and had an ASFR of 0.5. When we sum these ASFRs and multiply the total by 5, we have a cohort TFR of 1,986.5. This is the actual number of children that were produced on average by 1,000 members of the 1970–1974 fertility cohort.

A period (i.e., cross-sectional) TFR refers to the fertility of a **hypothetical cohort** (an imaginary set) of 1,000 women, whereas a cohort TFR refers to the actual fertility of a real cohort of 1,000 women. A period TFR for a year, say, the year 2007, refers to the fertility produced by women of all ages in the year 2007, a particular period of time. Alternately, a cohort

TFR refers to the fertility of a group of women who have already completed their childbearing years and sometimes may be viewed as out of date. Both period and cohort TFRs are used by demographers, but, as already noted, period TFRs are preferred principally because of their currency.

There are more fertility measures in addition to the four rates just covered, although none is as popular as the TFR. We now discuss two that are based on the concept of *population replacement*. Given a set of fertility rates, what does this mean with regard to replacing the population? Replacement refers to the production of female births, known among demographers as reproduction. Reproduction pertains to the production of female babies, while fertility refers to the production of babies of either sex.

The **gross reproduction rate (GRR)** is a standardized rate similar to the TFR, except that it is based on the sum of age-specific rates that include only female births in the numerators. Sometimes data are not readily available on the number of female live births reported by age of the mothers. Thus, the proportion of all births that are female is usually employed as a constant and is multiplied against the given TFR, as follows:

$$\text{GRR} = \text{TFR} * \left(\frac{\text{female births}}{\text{births}}\right) \tag{3.8}$$

The value of the constant multiplier, that is, the proportion of births that are female, varies only slightly from population to population. Most societies, but not all, have about 105 male babies born per 100 female babies. This results in about 51.2 percent of all births each year being male births and 48.8 percent being female births. Hence, we may use the following formula to calculate the GRR:

$$\text{GRR} = \text{TFR} * 0.488 \tag{3.9}$$

We noted previously that the TFR in the United States in 2005 was 2,055.2 (see Table 3.1). We may thus calculate the GRR for the United States in 2005 as follows:

$$\begin{aligned} \text{GRR} &= \text{TFR} * 0.488 \\ &= 2{,}055.2 * 0.488 = 1{,}002.9 \end{aligned} \tag{3.10}$$

This value of 1,002.9 may be interpreted as the number of daughters expected to be born alive to a hypothetical cohort of 1,000 women if none of the women died during their childbearing years and if the same schedule of age-specific rates applied throughout their childbearing years.

In Table 3.3, we show the calculation of the GRR for the United States in 2005 in greater detail than in formula 3.10. We start with a listing of the seven age groups (col. 1) and the midpoint of the age interval for each

Table 3.3. Calculation of gross reproduction rate (GRR) and net reproduction (NRR), United States

Age group	Midpoint of interval	Number of women in age group	Number of births to women in age group	Number of female births to women in age group	Female births per woman	Female birth rate during 5-year interval	Proportion of female babies surviving to midpoint of age interval	Surviving daughters per woman during 5-year interval
15–19	17.5	10,240,239	414,406	202,230	0.01974	0.09874	0.9889	0.09765
20–24	22.5	10,150,079	1,040,399	507,715	0.05002	0.25010	0.9848	0.24631
25–29	27.5	9,767,524	1,132,293	552,559	0.05657	0.28286	0.9801	0.27723
30–34	32.5	9,906,365	952,013	464,582	0.04690	0.23449	0.9752	0.22868
35–39	37.5	10,427,161	483,401	235,900	0.02262	0.11312	0.9691	0.10962
40–44	42.5	11,475,863	104,644	51,066	0.00445	0.02225	0.9600	0.02136
45–49*	47.5	11,372,141	6,546	3,194	0.00028	0.00140	0.9462	0.00133
						1.00296 = GRR		0.98218 = NRR

* Before 1997, data are for 45–49 per 1,000 women 45–49; beginning with 1997, rates are for live births to mothers 45–54 per 1,000 women 45–49.

Sources: For population data: U.S. Bureau of the Census, 2008; for birth data, Hamilton, Martin, and Ventura, 2007.

age group (col. 2). For instance, the midpoint for the age group 15–19 is 17.5. Data on the number of women in each age group in the United States in 2005 appear in column 3. Data on the number of babies born to the women in each age group appear in column 4.

As noted earlier, often we do not have data on the sex composition of births by age of mother, so this needs to be approximated. In the United States, as in most other countries, the **sex ratio at birth (SRB)** is around 105 boys per 100 girls. This means that the proportion of U.S. births that are female is 0.488. We multiply the birth data in column 4 by this constant proportion of 0.488, producing in column 5 the data on female births to women in each group. (If the SRB is not around 105, then another constant must be used. For instance, if we were calculating the GRR for China in 2000, we would need to take into account the fact that China's SRB in 2000 was 119. Thus, the proportion of all births in China in 2000 that were female was 0.457. This would be the constant used for calculating the GRR for China in 2000.)

Next, we divide the number of female births at each age (col. 5) by the number of women at each age (col. 3), to produce the age-specific rates of female births (col. 6); these age-specific rates have not yet been multiplied by 1,000.

The age-specific rates of female births in column 6 refer to only a single year, but we want to assume that the hypothetical cohort of women will experience these rates during the entire 5-year interval. So we now multiply the rates in column 6 by 5 to obtain the female birth rates during the 5-year interval; these are shown in column 7.

We then sum the values in column 7 to obtain the GRR. We multiply this sum by 1,000 to get the number of female births born to the hypothetical cohort of 1,000 women. The value of the GRR is 1.00296. We multiply it by 1,000, yielding 1,003.

The GRR makes no allowance for the fact that some of the mothers will die before they complete their childbearing years. Thus, to obtain a more accurate measure of the replacement of daughters by their mothers, we need another rate: the **net reproduction rate (NRR)**.

The NRR is a measure of the number of daughters who will be born to a hypothetical cohort of 1,000 mothers, taking into account the mortality of the mothers from the time of their birth. It may be thought of as the GRR net of mortality. Thus, the NRR subjects the 1,000 mothers not only to a schedule of age-specific reproduction rates but also to the risk of mortality up to the age of 49 (Rowland, 2003: 246). The formula for the NRR is as follows:

$$\mathrm{NRR} = \sum \left(\mathrm{ASFR}_{\mathrm{x\ to\ x+n}} * 0.488 * \frac{\mathrm{L_x}}{5\mathrm{l}_0} \right) \qquad (3.11)$$

The first two components of the NRR formula, namely, multiplying each ASFR by the constant of 0.488, are the major features of the GRR, as noted previously. We already know that the GRR refers to the number of female births born to this hypothetical cohort of 1,000 women if none of the women in the hypothetical cohort die from the time they are babies until the end of their reproductive years. The NRR takes into account the probabilities that not all of the women will survive from age 0 to the end of their childbearing years.

So, returning to Table 3.3, column 8 presents *life table values* for females of $\mathbf{L_x}$, multiplied by 500,000, for each of the age intervals. (The life table is discussed in detail in Chapter 5.) The $\mathbf{L_x}$ values represent the total number of person years lived in the age interval. Because we want the proportion of women who survive to the midpoint of the age interval, we divide each of the $\mathbf{L_x}$ values by $\mathbf{5(l_0)}$, or 500,000, giving us the proportion of women who survive from age 0 to the midpoint of each of the seven age intervals. To illustrate, for age interval 15–19, we take from the life table the value of $\mathbf{L_{15}}$ and divide it by $\mathbf{5(l_0)}$, or 500,000, equaling 0.9889. This means that 98.89 percent of the women in the hypothetical cohort will survive from birth through the midpoint of the age interval 15–19. These proportions of surviving females appear in column 8.

We then multiply the 5-year female birth rates (col. 7) by the proportions of surviving females (col. 8) to obtain in column 9 the number of daughters the women in each age group will bear, if they abide by a certain schedule of age-specific birth rates and a mortality schedule from the life table. We add the values in column 9 to get a total of 0.9822, which is the average net number of female births per woman in the hypothetical cohort after taking into account the mortality of the mothers; the net number of female births is less than one per woman. We multiply this figure by 1,000 for an estimate of the net number of female births for the entire cohort of 1,000 women, or 982.2 female births per 1,000 women.

The NRR is a "measure of how many daughters would replace 1,000 women if age-specific fertility and mortality rates remained constant indefinitely. Consequently, rates above 1,000 mean that eventually the population would increase, and rates below 1,000 mean that eventually the population would decrease, provided that the age-specific (birth and death) rates remained the same and no migration occurred" (Palmore and Gardner, 1994: 101–102).

PROXIMATE DETERMINANTS OF FERTILITY

Social demographers are primarily interested in ascertaining whether, how, and why various social, economic, cultural, and environmental factors influence both the likelihood of a woman having a baby and the number of babies she will have in her lifetime. Variables such as social class, economic status, religious beliefs, psychological disposition, attitudes about children, and many others have all been shown to be important in the decision to have a baby, as well as the number of babies (from zero to some positive number) a woman will have. Demographers have shown, for example, that the more years of education a woman has, the fewer will be her number of children. Why would this be? Why would more years of completed education result, on average, in a woman having fewer children? One reason is that women who are in school longer tend to marry later and also to start a family later, compared to women who have attended school for a shorter period of time. So, the education variable has an influence on fertility through the so-called proximate or intermediate variables that lie between – that is, they are intermediate to – the social, economic, cultural, and environmental variables and fertility.

In 1956, two famous demographers, Kingsley Davis and Judith Blake, wrote an influential paper about the behavioral and biological variables that are "intermediate" and thus directly influence fertility. These particular variables were distinguished from all of the other kinds of variables because the latter, by necessity, influence fertility by operating through the few intermediate variables specified by Davis and Blake. Briefly, the authors stated that the three variables of intercourse, conception, and gestation/**parturition** (the act or process of giving birth) should be considered as intermediate to the many other social, economic, cultural, and environmental factors that influence fertility. These latter variables, they argued, can only be related to fertility if they work through, or pass through, one or more of the intermediate variables.

Davis and Blake expanded on the three intermediate variables in the following ways: 1) the amount of intercourse is affected by the proportion of persons who marry, the length of time these persons are married, and their frequency of sexual intercourse while married; 2) the probability of conception is affected by **contraception** and by voluntary or involuntary **infecundity** (i.e., the inability to conceive); 3) the probability of a birth resulting from a given conception depends on the likelihood of miscarriage and abortion. They further emphasized that each intermediate variable can operate to increase as well as decrease fertility. Not all of them need to operate in the same direction, however. The observed level of fertility in a population depends on the net balance of *all* the intermediate variables.

Let us pause for a moment and mention some of the fertility terms to be used here. We already know the difference between fertility and fecundity. *Fertility* refers to actual reproduction and *fecundity* refers to the ability to reproduce. The opposite terms are *infertility* (also called **childlessness**) and *infecundity* (which is synonymous with **sterility**). Sterility implies the existence of infertility, but the reverse is not necessarily the case. A **fecund** woman may choose to remain infertile by not marrying or by practicing highly effective contraception. Infertility, then, is due to a voluntary decision not to have children or it is caused by (biological) infecundity.

With regard to fecundity, it is sometimes divided into five categories on the basis of the extent of fecundity impairment and the degree of certainty of the evidence (Poston, Zhang, and Terrell, 2008). That is, how able or unable are couples with respect to producing a birth?

First, all couples may be classified as either "fecund" or "subfecund." Fecund couples are capable of giving birth. Second, subfecund couples have impairments of one sort or another and may be subdivided into the following groups: definitely sterile, probably sterile, **semifecund**, and **fecundity indeterminate** (Badenhorst and Higgins, 1962). Definitely sterile couples are those for whom conception is impossible because of certain physical or medical conditions, including an operation, some other impairment, or menopause. Probably sterile couples are those for whom a birth is improbable on the basis of specific medical evidence. Semifecund couples are those who have married or cohabited for a relatively long time without using contraception but have not conceived. Fecundity indeterminate couples are those who meet the criteria for semifecund couples, except that the woman sometimes reports douching "for cleanness only" soon after intercourse. These couples are defined as "fecundity indeterminate" (Badenhorst and Higgins, 1962: 281). Demographic research has shown that the majority of subfecund couples are impaired according to these definitions (Poston, Zhang, and Terrell, 2008).

Returning to our discussion of the proximate determinants, the demographer John Bongaarts, in papers written in 1978 and 1982, respecified the intermediate variables of Davis and Blake in a way that facilitated the quantitative specification of how they influence fertility. He referred to the intermediate variables as the "proximate determinants" of fertility, a term very commonly used these days. The proximate-determinants perspective is one of the most elegant and useful frameworks in demography for understanding the process of fertility and how it is determined.

Bongaarts recognized that the "Davis and Blake framework for analyzing the determinants of fertility has found wide acceptance, [but] it has proven difficult to incorporate into quantitative reproductive models" (1982: 179). He thus set out the following seven proximate determinants: 1) marriage and marital disruption, 2) contraceptive use and effectiveness, 3) prevalence of induced abortion, 4) duration of postpartum infecundability, 5) waiting time to conception, 6) risk of intrauterine mortality, and 7) onset of permanent sterility. Let us consider each of these in turn.

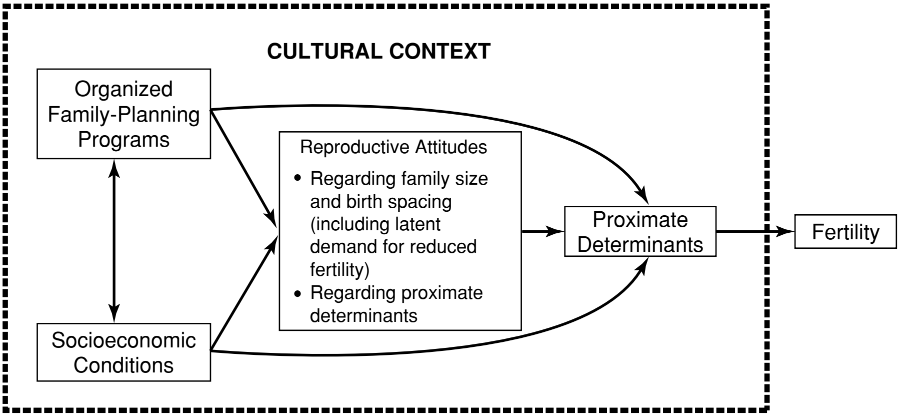

Figure 3.3. The determinants of fertility. *Source:* Knodel, Chamratrithirong, and Debavalya, 1987: 11.

First, with regard to marriage and marital disruption, this variable was intended to reflect the proportion of women in the population who were of reproductive age and engaging in sexual activity on a regular basis. Second, the contraception variable reflects deliberate contraceptive behaviors including abstinence and sterilization that are undertaken to reduce the risk of conception. Third, induced abortion includes all practices and behaviors that deliberately interrupt pregnancies. Fourth, duration of postpartum infecundability, in Bongaarts's terms, may be considered this way: "Following a pregnancy a woman remains infecundable (i.e., unable to conceive) until the normal pattern of ovulation and menstruation is restored. The duration of the period of infecundity is a function of the duration and intensity of lactation" (1978: 107). Fifth, waiting time to conception refers to the fact that a woman who is not contracepting is able to conceive "for only a short period of approximately two days in the middle of the menstrual cycle when ovulation takes place. The duration of this fertile period is a function of the duration of the viability of the sperm and ovum" (Bongaarts, 1978: 107). Sixth, risk of intrauterine mortality reflects the fact that many conceptions do not eventually lead to live births because some of them end as miscarriages, spontaneous abortions, or stillbirths. Seventh, the onset of permanent sterility reflects the fact that "women are sterile before menarche . . . and after menopause, but a couple may become sterile before the woman reaches menopause for reasons other than contraceptive sterilization" (Bongaarts, 1978: 107–108). (**Menarche** is the beginning of the female reproductive period, signaled by the first menstrual flow, and **menopause** is the end of that period, signaled by the termination of menstruation.)

Figure 3.3 summarizes the broad framework incorporating the proximate determinants that demographers use to study fertility. The figure highlights three major kinds of variables that occur in time prior to the proximate determinants: **family planning** variables, socioeconomic variables, and attitudinal variables. All three types of variables must work through the proximate determinants to influence fertility. Moreover, as John Knodel and his colleagues have observed, "since many of the proximate determinants are influenced by volitional actions and choice, attitudes directly concerning fertility as well as attitudes about the proximate determinants themselves are an important feature of any exposition of reproductive change" (Knodel, Chamratrithirong, and Debavalya, 1987: 9).

Bongaarts (1982) has taken the first four of the proximate determinants and quantified them with indices ranging from 0 to 1, with the lowest value of 0 indicating the greatest possible inhibiting effect on fertility of the proximate determinant and the maximum score of 1 representing no inhibiting effect of the determinant. The marriage-pattern index, *Cm,* has a value of 1 when all women of reproductive ages are in a marital or consensual union and 0 when none of them is in such a union. The contraception index, *Cc,* equals 1 if no contraception is used in the population and equals 0 if all fecund women are using completely effective modern methods of contraception. The postpartum-infecundability measure, *Ci,* is an index that ranges from a value of 1 when no women are experiencing postpartum infecundability to a value of 0 when all women are. The index of abortion, *Ca,* ranges from a maximum value equaling 1 when there is no induced abortion practiced in the population to 0 if every pregnancy that occurs is aborted.

Bongaarts (1982) conducted a quantitative analysis of the fertility of 41 historical and contemporary (developing and developed) countries and populations and showed that 96 percent of the variation in fertility could be explained solely by variation in the four proximate determinants of marriage, contraceptive use, postpartum infecundability, and abortion (see also Bongaarts and Potter [1983]). The other three of the proximate determinants were less important and did not vary significantly among the populations. Bongaarts's demonstration is very important because it shows that virtually all of the variation in fertility is due to variation in only the four major proximate determinants. Thus, the effects of any and all variables to the left of the proximate determinants (see Figure 3.3) can only have an effect on fertility if they operate through the proximate determinants.

THEORIES OF FERTILITY

Demographers have developed several different theories of fertility. Prominent explanations are **demographic transition theory**, **wealth flows theory, human ecological theory**, political economic theory, feminist theory, proximate determinants theory, bio-social theory, relative income theory, and diffusion theory (Poston and Terrell, 2006). As we have already noted, there is an abundance of theory in demography, indeed, more than in most of the social sciences.

A major explanation of fertility change and dynamics has its origins in demographic transition theory (DTT), as first developed by Warren S. Thompson (1929) and Frank W. Notestein (1945). We discuss DTT in more detail in Chapter 9; it is an important perspective for describing changes in fertility.

Current versions of DTT propose four stages of fertility (and mortality) decline that occur in the process of societal modernization. The first stage is the preindustrialization era, with high birth and death rates and **stable population** growth. (A stable population is a hypothetical population with unchanging birth, death, and growth rates and age composition.) With the onset of industrialization and modernization, the society transitions to lower death rates, especially lower infant and maternal mortality, but maintains higher birth rates with the result of rapid population growth. The next stage is one of decreasing population growth due to lower birth and death rates, which leads then to the final stage of low and stable population growth.

DTT argues that the first stage hinges on population survival. High fertility is necessary because mortality is high. Thus, societies develop a variety of beliefs and practices that support high reproduction, which are centered primarily on the family and kinship systems. The forces of modernization and industrialization alter this state of near-equilibrium, and the first effect is often a reduction in mortality. Indeed, the beginnings of mortality decline in many European countries were stimulated not so much by medical and public health improvements as by a general improvement in levels of living. This intermediate stage resulted in rapid rates of population growth because fertility remained high after mortality had declined. In the next stage, fertility also declines to lower levels. The causal linkages are complex. Underlying the global concepts of industrialization and modernization are specific determinants of fertility such as women's participation in the labor force and the changing role of the family. The normative, institutional, and economic supports for the large family become eroded, and the small family becomes predominant. The increasing importance of urbanization affects the family by altering its role in production. Also, urban families have considerably higher demands for consumption by their children, especially for education and recreation (Browning and Poston, 1980; Coale and Watkins, 1986; Davis, 1963; Hirschman, 1994; Knodel and van de Walle, 1979; McKeown, 1976; Poston, 2000; Poston and Terrell, 2006).

John C. Caldwell (1976) has revised DTT with his fertility theory of wealth flows. It is based on the notion that the "emotional" nucleation of the family is crucial for lower fertility. This occurs when parents become less concerned with ancestors and extended family relatives than they are with their children, their children's future, and even the future of their children's children (1976: 322). Caldwell explained that this depends on the direction of the intergenerational flows of wealth and services. If the flows run from children to their parents, parents will want to have large families. In modern societies where the flow is from parents to children, they want small families or maybe even no children (Poston and Terrell, 2006; Poston, Zhang, and Terrell, 2008).

Two other prominent theories of fertility change are based on human ecology and political economy. Both are extensions of demographic transition theory but in different ways. Human ecological theory is a macro-level explanation and argues that the level of sustenance-organization complexity of a society is negatively related with fertility growth and decline (Poston and Frisbie, 2005). In the first place, high-fertility patterns are dysfunctional for an increasingly complex sustenance organization because so much of the sustenance produced must be consumed directly by the population. High fertility will reduce the absolute amount of uncommitted sustenance resources, thereby limiting the population's flexibility for adapting to environmental, technological, and other kinds of changes and fluctuations. Low fertility is more consonant with the needs and requirements of an expansive sustenance organization. More sustenance would be available for investment back into the system in a low-growth and low-fertility population than in a population with high fertility. Hence, large quantities of sustenance normally consumed by the familial and educational institutions in a high-fertility population would be available as mobile or fluid resources in a low-fertility population. Sustenance organization in this latter instance would thus have the investment resources available for increasing complexity, given requisite changes in the environment and technology. This leads to the hypothesis of a negative relation between organizational complexity and fertility and population change (Kasarda, 1971; London, 1987; London and Hadden, 1989; Poston and Frisbie, 2005).

The political economic approach is another way to analyze fertility. Diverse fields of knowledge are integrated into the political economy approach so that research reflecting this perspective is "multileveled," combining both macro- and micro-level explanations. Its complexity requires a methodology that embraces both quantitative and qualitative approaches (Greenhalgh, 2008; Poston and Terrell, 2006).

The political economy of fertility is not really a theory of fertility per se but an investigative framework, or "analytic perspective," for the study of fertility (Greenhalgh, 1990b: 87). A good example of a political economy approach to fertility is David I. Kertzer and Dennis P. Hogan's 1989 study of Casalecchio, Italy. The authors tracked one small, rural Italian community during a few change-laden decades of the nineteenth and twentieth centuries, using individual-level data and directed by a life-course perspective. They touched on historical events, such as labor and marriage patterns, often ignored by other studies of demographic change. They showed that fertility rates and fertility reduction vary depending on the class or occupation of the family, thus demonstrating that macro-level socioeconomic factors have idiosyncratic effects on different classes of people.

WORLD FERTILITY TRENDS AND PATTERNS

In the world in 2005, the total fertility rate (TFR) was 2.65, but this value hides the tremendous heterogeneity in fertility in the countries around the world (United Nations, 2005). As of 2005, there were 205 countries; a country is an internationally recognized country or a territory with a population of 150,000 or more; or, if smaller than 150,000, is a member of the United Nations (UN). Of these 205 countries, 73 had TFRs at or below the replacement level of 2.1. (Since in most developed countries around 97 to 98 percent of babies survive to become parents, 1,000 females need to give birth to 2,100 babies in order for 2,000 of them to live to become parents.) Of the remaining countries, 35 had TFRs of 5.0 or higher; of these, 30 of the countries are considered by the UN to be the least developed of all countries and are located mainly in Africa (in the countries of Ethiopia, Malawi, Rwanda, Somalia, Uganda, Angola, and Chad, among others) (United Nations, 2005: 6).

The countries with low fertility fall roughly into three groups. The first includes most of the countries in northern and western Europe, as well as Argentina, Australia, Canada, New Zealand, and the United States. These countries had declining fertility rates in the twentieth century until the 1930s, and then their fertility either leveled off or declined, except for some like the United States and many countries in Western Europe where baby booms occurred. The second group of low-fertility countries consists of those in southern and eastern Europe. These nations had fairly consistent fertility declines in the 1930s, 1940s, and 1950s. The third group includes South Korea, China, Japan, Taiwan, and Singapore, which had relatively high fertility rates until the end of World War II and into the 1960s but sharp declines thereafter (United Nations, 2005: 6).

Birth rates in some of the countries of the developing world are high but not as high as biologically possible. CBRs have probably never been higher than 65 to 70 in any countries of the world (Kuczynski, 1936).

A country with a CBR of 70 would have a TFR of roughly 9.5. In prerevolutionary Russia, the birth rate in several provinces was around this level (United Nations, 1973: 73). Fertility rates approaching this level are possible only when almost everyone is engaged in frequent sexual intercourse, there is little if any contraception, childbearing begins at a young age, and families have many children. There are documented cases of women having twenty or more births. However, the largest *average* family size that seems possible ranges from around nine to thirteen children, and then only in societies where marriage occurs at an early age and birth prevention is not practiced. For instance, rural Irish women who married under age 20 and who had completed childbearing by the date of the 1911 census averaged 8.8 live births (Glass and Grebenik, 1954). T. E. Smith (1960) has reported that Cocos Island women marrying at age 14 or 15 averaged 10.8 children by age 45. Among the Hutterites, a Protestant religious sect living in small agricultural settlements in the Dakotas, Montana, and the adjacent Canadian provinces of Manitoba and Saskatchewan, there was in the 1950s an average of around twelve confinements among women whose last child was born when they were 45 or older (Tietze, 1957). These very high birth rates, however, fall far short of the biological maximum (Bongaarts, 1975).

As of around 2005, the highest TFRs in the world were 7.3 in the Democratic Republic of the Congo and 7.1 in Niger. The lowest TFRs were 0.8 in Macau; 1.0 in Hong Kong; and 1.2 in Ukraine, the Czech Republic, Slovakia, and South Korea. Figure 3.4 shows all the countries of the world ranked by the values of their TFRs in around 2005, high to low, left to right. There is a tremendous range in fertility in the countries, certainly as large a range as ever experienced in the history of the world.

Despite the very high fertility rates in many of the developing countries, some are fairly far along in the transition from high birth rates to low birth rates. However, in fifteen of the countries, "there is either no recent evidence about fertility trends or the available evidence does not indicate the onset of a fertility reduction. Although [their fertility] is projected to decline after 2010 at a pace of about one child per decade, none is expected to reach 2.1 children per woman by 2050" (United Nations, 2005: 7). Most of these countries – namely, Afghanistan, Angola, Burkina Faso, Burundi, Chad, Congo, Democratic Republic of the Congo, Democratic Republic of Timor-Leste, Guinea-Bissau, Liberia, Malawi, Mali, Niger, Sierra Leone, Somalia, Uganda, and Yemen – are classified by the UN as "least developed." Many have been significantly affected by the **human immunodeficiency virus (HIV)**/AIDS epidemic, and some have experienced "civil strife and political instability in recent years, factors that militate against the

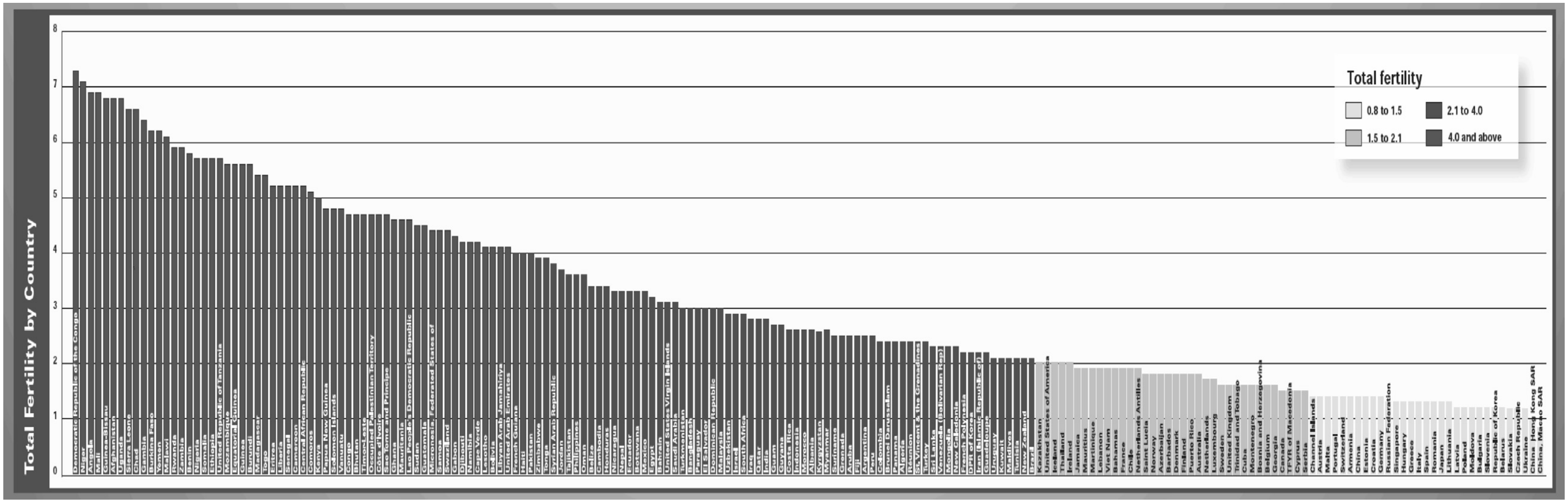

Figure 3.4. Total fertility rates, countries of the world, circa 2005. *Source:* United Nations, 2008a.

provision of basic services for the population" (United Nations, 2005: 8). Certainly, these considerations stand in the way of demographic advances in future years and are serious challenges to the future development of these countries.

As already noted, fertility levels in 73 countries had all declined to replacement levels or below by around 2005. In the 1970s, there were only eighteen countries with fertility below replacement. These included the European countries of Hungary, Denmark, Finland, Sweden, Germany, and Switzerland. In almost all them, fertility continued to decline, attaining "historically unprecedented low levels (below 1.3 children per woman) in fifteen developed countries, all located in Southern and Eastern Europe" (United Nations, 2005: 9; see also Kohler, Billari, and Ortega, 2002; and Morgan, 2003).

Following Francesco C. Billari (2004), we refer to fertility as being "low" when the TFR is below the replacement level of 2.1, as being "very low" when it is below 1.5, and as being "lowest low" when it is below 1.3. In 2005, eleven countries reported lowest-low fertility – that is, a TFR below 1.3; these include South Korea, Taiwan, Poland, the Czech Republic, Slovakia, and Ukraine.

No discussion of world fertility trends would be complete without mentioning depopulation, that is, the decline in the size of the population. This is so because depopulation is projected to occur in most countries of the world in the next fifty to a hundred years. Despite the vast amount of attention paid since the late 1960s to the phenomenon of overpopulation (see esp. Ehrlich [1968]; B. Friedman [2005]; and Meadows, Randers, and Meadows [2004]), declines in population are expected to occur in around fifty or more countries by the year 2050 (Population Reference Bureau, 2006) and in even more countries thereafter (Howden and Poston, 2008).

Even though the population of the world is projected to continue to grow, reaching around 9.1 billion in 2050 according to the United Nation's medium-variant projection, a slowing of the rate of population growth is already underway, and a decline in the size of the world population could begin as early as 2050 (United Nations, 2005). The region most significantly impacted by depopulation is Europe. Between 2000 and 2005, at least sixteen countries in Europe experienced a decline in population size. The largest net loss occurred in Russia, with a loss of almost 3.4 million persons in the five years between 2000 and 2005 (United Nations, 2005).

Indeed, Russia's population is expected to drop from 143 million in 2005 to around 124 million by 2030. Nicholas Eberstadt (2009: 51) has noted the severity of this depopulation in Russia in his statement that

"Russia's human numbers have been progressively dwindling. This slow motion process now taking place in the country carries with it grim and potentially disastrous implications that threaten to recast the contours of life and society in Russia, to diminish the prospects for Russian economic development, and to affect Russia's potential influence on the world stage in the years ahead."

For the majority of countries, including Russia, the reason for depopulation is sustained low fertility. For a population to remain stable, TFRs need to be at or below replacement, that is, no higher than 2.1 children per woman, and the cohorts in the childbearing ages cannot be larger than those in other age groups. If there are large numbers in the population in the parental ages, **replacement-level fertility** alone will not result in depopulation. This is due to what is referred to as negative **population momentum**, that is, the lag between the decline in TFRs and the decline in CBRs that is caused by large numbers of women still in their childbearing years owing to past high fertility. In the 2000–2005 period, the UN noted that sixty-five countries had fertility rates below replacement levels, with fifteen at extremely low levels (i.e., a TFR below 1.3) (United Nations, 2005). In 2006, the Population Reference Bureau reported that as many as seventy-three countries were experiencing TFRs below the replacement level (see also Howden and Poston, 2008).

Many countries with relatively high fertility have started to experience declines in their fertility. The UN has observed that of the thirty-five countries with a TFR of 5 or more, twenty-two have experienced declines in fertility in the past ten to fifteen years (United Nations, 2005). These lower rates of fertility, coupled with low rates of mortality and immigration, are responsible for depopulation in the majority of countries projected to lose population in the next fifty years. For a few countries, population decline is expected to occur even though their fertility is greater than the replacement levels. These countries – namely, Botswana, Lesotho, and Swaziland – are being significantly impacted by the HIV/AIDS epidemic, leading to a net loss in population (Howden and Poston, 2008).

The depopulation of most of the countries in the developed world has significant economic impacts and implications. Major effects will likely be felt through the **aging of a population**. As fertility declines, birth cohorts become progressively smaller. These smaller birth cohorts, coupled with increases in life expectancy, lead to an increasingly larger proportion of the population that is older than age 65 and a smaller proportion of the population in the working ages. The UN has reported that the period between 2005 and 2050 will see a doubling of the **old-age dependency ratio** (**ADR**) (i.e., the ratio of the population aged 65 and older to the population aged 15–64, times 100) in developed countries from 22.6 to

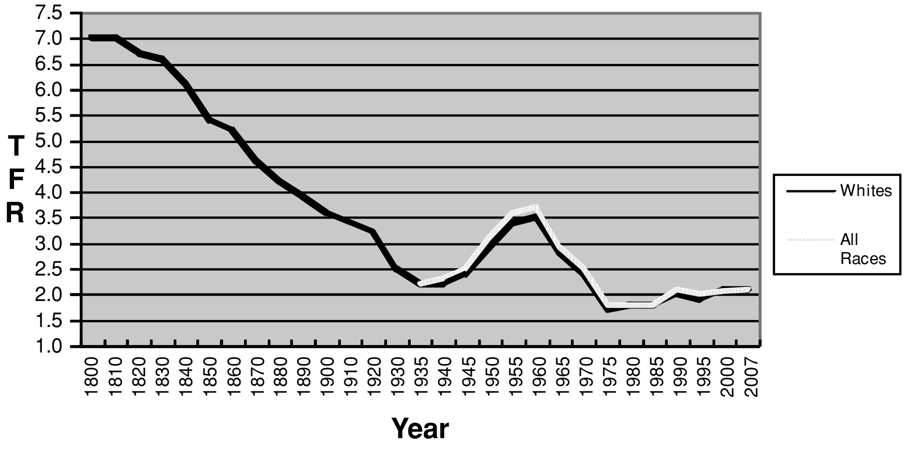

Figure 3.5. Total fertility rates, United States 1800–2007.

44.4 (United Nations, 2005). For many countries, health care and pension programs are ill equipped to handle the large increases in the numbers of elderly, who themselves will live longer than their predecessors (Howden and Poston, 2008).

FERTILITY CHANGE IN THE UNITED STATES

In less than 220 years, the United States has increased tremendously in size from fewer than 4 million people in 1790 to 300 million in 2006 (see Chapter 10 for more discussion). Figure 3.5 shows TFRs for the United States from 1800 to 2007. TFRs for whites are shown for all of the years, and TFRs for all races combined are reported beginning in 1935.

A high fertility rate was an important component of the rapid population increases in the United States in the early years between 1790 and 1860. Indeed, in 1800, the white population had a TFR of 7.0, which was likely the highest fertility rate of any country in the world at the time (Haines and Guest, 2008; Sanderson, 1979). The TFR in 1810 was still around 7.0 but by 1820 had dropped slightly to 6.7. It declined to 5.2 in 1860 and then to just under four births per woman by 1900 (Hamilton, Martin, and Ventura, 2007; Taeuber and Taeuber, 1958; U.S. Bureau of the Census, 1975).

High fertility in the United States resulted from the fact that more than one-half of the population was fecund. The average age of the population was 15.9 in 1790, 17.0 in 1820, and 19.4 in 1860. Another reason for high fertility was the high levels of rural and agricultural activity. When the first census of the United States was conducted in 1790, 95 percent of the population was rural. The rural portion comprised about 90 percent of the population through 1850 and dropped to only 75 percent by 1900 (Kahn, 1974).

Between 1790 and 1860, when agriculture was dominant, income was directly related to the acreage of cultivated fields. Fields in the West were cheap and easy to obtain, and many people tended to have more children in order to have more laborers. Big families were popular and normative.

Every modern, economically developed nation has undergone a demographic transition from high to low levels of fertility and mortality. The United States experienced a sustained fertility decline starting in the nineteenth century (Sternlieb and Hughes, 1978). The TFR (for whites) dropped to 4.6 in 1870 and to 3.6 in 1900 (U.S. Bureau of the Census, 1975). By 1920, just after World War I, the white TFR had dropped to 3.2 (see Figure 3.5).

In the several decades of the early twentieth century, the United States experienced a rapid transition in its economy, industrialization, and urbanization (Taeuber and Taeuber, 1958). In the late 1950s at the height of the baby boom era, the TFR reached its peak at 3.7. The high (baby boom) fertility following World War II was promoted in part by a need to take into account the population losses that occurred during the war.

In the 1960s, the fertility rate started to decline. Many factors influenced this reduction, such as higher living expenses, increases in educational opportunities and expectations, and more women employed in the labor force. The average U.S. family size was reduced. Cheap, easily accessible, and more effective contraceptives, along with abortions (see discussion in Chapter 4), gave couples greater control over births and hence were another factor in the fertility decline. In 1972, the U.S. TFR dropped for the first time below the replacement level of 2.1 (Kahn, 1974); by 1975, it was 1.7. Even during the Great Depression years when fertility was low, it never dropped below 2.2. Total population increase in 1972 was only 0.7 percent, almost one-half of the average annual increase during the 1960s (Kahn, 1974). U.S. birth rates kept declining, although not as rapidly as in earlier years. Since 1990, the TFR has remained at just above two children per woman (see Figure 3.5).

The U.S. population of 281 million people in 2000 has been growing slowly, at a rate of around 1 percent a year. The TFR of 2.05 births in each woman's lifetime is just below the replacement level of 2.1 but, nevertheless, is the highest of any of the developed countries in the world. If these trends continue, the American population is projected to level off at about 408 million by the middle of this century and then begin to decline, even allowing for substantial immigration. People are living longer, and the U.S. population as a whole is aging.

ADOLESCENT FERTILITY

Adolescent fertility refers to the childbearing of young women. The **adolescent fertility rate** is represented by the ASFR for women aged 15–19. Fertility rates for young women in this age group are a particularly important indicator of women's status because women who give birth at this young age "often forego the opportunity to study or find employment away from home" (United Nations, 2005: 1). The adolescent fertility rate for the world for the period 2000–2005 was 55 per 1,000. That is, in 2000–2005, on average each year, for every 1,000 young girls of ages 15–19 in the world, there were 55 babies born to them. Among the developed countries, the adolescent fertility rate was 24, varying from a low of 5 in Belgium and Switzerland to a high of 43 in the United States. Only the United States and two other countries in the developed world have adolescent fertility rates above 30, namely, Bulgaria (40) and Romania (33). In the developing countries, the adolescent fertility rate ranges from lows of 2 in South Korea and 3 in China to highs of 199 in Niger and 192 in Mali. In South Korea and China, for every 1,000 adolescent women, only two or three had babies in an average year in the 2000–2005 period. Almost twenty-two times as many adolescent women in the United States gave birth as in China or South Korea. In both Niger and Mali, almost one in five adolescents gave birth during this period (United Nations, 2007). There is more variation among the countries of the world in adolescent fertility than in fertility per se (as measured by the TFR).

The UN data published in 2007 indicate that the adolescent fertility rate is above 90 in at least fifty-five developing countries; of these, thirty-five are in Africa, fifteen in Latin America, and seven in Asia. The rate is 30 or lower in thirty-two developing countries. It is interesting that the so-called developing countries have the highest and the lowest adolescent fertility rates in the world.

We noted that the United States has by far the highest adolescent fertility rate of all the countries of the developed world: 43 births per 1,000 adolescents, calculated for the year 2002. But adolescent fertility for U.S. females varies considerably by race and ethnic group. The lowest adolescent fertility rate for U.S. women in 2002 was for Anglo females, 28/1,000. The highest was for Hispanic women, 83/1,000. Black adolescents, American Indian or Alaskan Native adolescents, and Asian or Pacific Islander adolescents had fertility rates between these two extremes at 68/1,000, 54/1,000, and 18/1,000, respectively. The very high adolescent fertility rate of U.S.

Hispanics in 2002 was about equal to the rates of Panama, Venezuela, and the African countries of Namibia, Togo, and Mauritania.

The high Hispanic adolescent fertility rate applies only to Hispanics of Mexican origin. (Cubans have lower rates of adolescent fertility.) Current research indicates that young women in the United States of Mexican origin have particularly high rates of adolescent fertility mainly because of the barriers that limit their access to higher education (Conde, forthcoming).

MALE FERTILITY

An important limitation of the our discussion of fertility theories and measures is that they are all derived from calculations and analyses of female fertility. The fertility of men and the determinants of male fertility have rarely been examined and compared with those of females, but they should be. Males are a neglected minority in fertility studies.

Several reasons have been proposed by demographers to justify the exclusion of males from fertility studies. The biological reasons are that the fecundity and the childbearing years of women occur in a more sharply defined and narrower range (15–49) than they do for men (15–79), and that "both the spacing and number of children are less subject to variation among women; a woman can have children only at intervals of 1 or 2 years, whereas a man can have hundreds" (Keyfitz, 1977a: 114). Among the methodological reasons are that data on parental age at the birth of a child are more frequently collected on birth-registration certificates for the mothers than for the fathers, and that when such data are obtained for mothers and fathers, there are more instances of unreported age data for fathers, especially for births occurring outside of marriage. The sociological reasons include the fact that men are regarded principally as breadwinners and "as typically uninvolved in fertility except to impregnate women and to stand in the way of their contraceptive use" (Greene and Biddlecom, 2000: 83; Poston, Zhang, and Terrell, 2008).

In Texas in 2000, among birth certificates that listed information on the father's age (85.4 percent of all birth certificates), the ages of the fathers ranged from a low of 13 to a high of 80, with a mean of age 29.2.

Biological and demographic studies provide us with evidence of the importance of men in fertility and related behaviors. Biologists have found that the male sex in most all species contributes an equivalent amount of genetic information to the next generation as does the female. But the variance contributed by the male sex to the next generation is often greater than that of the female sex, especially in species where **polygyny** (a union where a male is simultaneously married to two or more females) is practiced. That is, most females reproduce, some males do not reproduce, and other males have a large number of offspring. This pattern has been found in most mammalian species (Coleman, 2000). In addition, it has been found that there are more childless males than females in many species.

Demographically, it has been shown that men have different patterns of fertility and fertility-related behavior. Males' age-specific fertility starts later, stops much later, and is typically higher than that of females (Paget and Timaeus, 1994). The TFRs for males and females are not identical either. Research has shown that in most industrialized countries, male TFRs are higher than female TFRs (Coleman, 2000). In the 1990s and later, countries with both male and female TFRs lower than 2.2 tended to have more similar than dissimilar male and female TFRs, and the opposite was found for countries with male and female TFRs higher than 2.2 (Zhang, 2007). This means that for men and women, if they followed the sex- and age-specific fertility rates of an area at one point in time, and if they had fewer than 2.2 children during their childbearing years, these men and women were more likely to have similar than dissimilar fertility and vice versa (also see Myers, 1941; and D. Smith, 1992). (An **age/sex-specific rate** refers to the demographic behavior – for example, regarding fertility – of a subset of the population categorized by age and sex.)

The special importance of male fertility is also seen in the determinants of fertility and fertility-related behaviors, such as cohabitation and marriage. For example, in the United States, men's fertility is more likely to be influenced by their marital and employment status compared to women's. Being married and employed significantly increases men's number of children ever born (CEB), while such factors do not have as strong an impact on women (Zhang, 2007). The relevance of labor-force participation for women's fertility has been emphasized repeatedly. In our own research, we have used several independent variables from various fertility paradigms, namely, human ecology, political economy, and wealth flows, to predict both male and female TFRs for the counties of Taiwan. The variables have consistently performed better when predicting variation in female TFRs than in male TFRs (Poston, Baumle, and Micklin, 2005).

Men also have different cohabitation and marriage patterns compared to women. In the United States, for birth cohorts born during 1958 to 1987, living alone, being foreign-born, and living in fragmented families all tend to increase the likelihood of cohabitation for women but not for men. Foreign-born men are more likely to marry than native-born men. But these factors do not have as significant an impact on women's marriage behavior (Zhang, 2007). Researchers have conducted studies examining male and female transitions to adulthood in twenty-four European countries using survey data for the 1980s and 1990s. They have found that, in general, the negative effect on fertility of educational attainment is stronger for women than for men. Also, unemployment leads to men's postponement of marriage, whereas it affects women in two distinct ways: It either accelerates or slows down the timing of marriage for women. The effect of religion is stronger among women than among men. Furthermore, being Catholic and attending church services affect men's and women's parenthood timing in different ways in Catholic-prevalent countries. Other relevant factors such as parental influence seem to have a different impact for males than for females (Corijn and Klijzing, 2001).

Historically, women have been tied to motherhood, and this is deeply rooted in law and policy in the ways that jobs are structured and that family relations are navigated. Studies of fertility and parenthood have been undertaken by demographers in a similar way (Riley, 2005). These, together with the biological and methodological reasons stated previously, have resulted in the decreased attention given to males in fertility research. It was mentioned earlier that biological and demographic analyses and results have shown that fertility and parenting are not simply female issues; they are issues involving both men and women. The study of fertility should not focus only on females. Greenhalgh (1990b) and Riley (1998; 2005) encouraged more discussion about gender issues among demographers, and critical demography has promoted bringing men into population studies (Coleman, 2000; Horton, 1999). It is necessary to incorporate gender studies and other disciplines into studies of demography in order to gain a more balanced picture of demographic issues. Indeed, male fertility is one of the emerging issues of demographic study. Demographers and sociologists need to give more attention to males in their analyses of fertility variation and change. It is essential to take men's roles and commitments into account when considering factors leading to decisions about the bearing and rearing of children (Poston, Baumle, and Micklin, 2005; Poston, Zhang, and Terrell, 2008).

CONCLUSION

In this chapter, we first considered various measures of fertility and next discussed the so-called proximate determinants of fertility. These are the mainly biological factors that lead directly to fertility and that are influenced heavily by social factors. Indeed, the various social, economic, cultural, environmental, and psychological factors that affect fertility do so only through the "proximate" variables. Both the societal birth rate and the fertility of individual women and men are produced by a combination of those factors. We next looked at some of the main theories generated by demographers to account for the reasons that some women or men or societies have more babies than other women or men or societies. We then considered world fertility patterns and U.S. fertility trends and differences. We concluded with a discussion of adolescent fertility and male fertility.

We close this chapter with a summary of some of the major fertility differentials. Fertility rates are much higher in many developing countries, especially those in sub-Saharan Africa, than they are in developed countries. Likewise, different types of people have different patterns of fertility. Generally, the higher a person's socioeconomic status, the fewer children that person is likely to have. In industrialized societies, women employed in the labor force tend to have fewer children than women who are not so employed. Having a smaller family also increases the woman's availability for employment, and her employment per se encourages a small family. Levels of childbearing also tend to be lower in urban than in rural areas. This is particularly true in more modernized countries, although in the past three or four decades, the difference between rural and urban childbearing rates has diminished. There are few if any differences in childbearing between Catholics and non-Catholics in the United States and other more developed countries. These differences are due mainly to differences in socioeconomic status. The fertility of Moslem women, however, is higher than that of Christian and Jewish women, both in developed and developing countries. These differentials notwithstanding, the most important point to remember about fertility is that fertility rates are heavily conditioned by social, economic, psychological, cultural, and environmental factors, all of which eventually affect fertility through the proximate determinants.

KEY TERMS

adolescent fertility rate
age/sex-specific rate
age-specific fertility rate (ASFR)
aging of a population
at-risk population
childlessness
cohort analysis
conception
contraception
crude birth rate (CBR)
demographic transition theory (DTT)
family planning
fecund
fecundity
fecundity indeterminate
fertility
general fertility rate (GFR)
gross reproduction rate (GRR)
human immunodeficiency virus (HIV)
human ecological theory
hypothetical cohort
infecundity
less-developed countries (LDCs)
menarche
menopause
net reproduction rate (NRR)
old-age dependency ratio
parturition
period perspective
period rate
polygyny
population momentum

proximate determinants
reproduction
replacement-level fertility
semifecund
sex ratio at birth (SRB)
specific rate
stable population
sterility
total fertility rate (TFR)
wealth flows theory
zygote